

FITTRACK

AI-Based Fitness Tracking and Personalized Fitness Recommendations

Project Documentation

Submitted by

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1. Introduction

FITTRACK is a secure, AI-powered fitness tracking application that helps users maintain their workout records and receive personalized fitness recommendations. The system provides RESTful APIs for user authentication, workout management, workout searching and AI-assisted fitness guidance.

The backend is developed using Node.js and Express.js, and MongoDB is used to store user and workout information through Mongoose. FITTRACK follows a modular MVC architecture and uses JWT authentication to protect user-specific operations, while passwords are handled securely using bcrypt.js. The system also integrates Google Gemini AI to generate personalized workout recommendations and fitness insights based on user information and workout statistics. Centralized error handling ensures proper validation and consistent management of application errors.

2. Scenario-Based Case Study

2.1 Background

Rahul is a fitness-conscious user who regularly performs workouts and wants to maintain a record of his daily fitness activities. Initially, he records his workouts manually. As the number of workouts increases, he finds it difficult to organize his workout history, track his progress and decide which workouts are suitable for his fitness goals. He also wants a system that provides personalized suggestions instead of searching manually for suitable exercises.

2.2 Problem

The manual approach creates the following difficulties:

- Difficulty maintaining and searching workout history.
- No centralized fitness record.
- No secure authentication.
- No personalized workout recommendations.
- Difficulty understanding workout performance due to manual calculation and interpretation of workout statistics.

2.3 Solution

FITTRACK provides a centralized backend that allows users to register and log in securely, store, view, search, update and delete workout records, and generate personalized workout recommendations.

3. System Requirements

3.1 Software Requirements

Requirement	Specification
Operating System	Windows 10/11, macOS or Linux
Runtime	Node.js v16+
Package Manager	npm v8+
Backend Framework	Express.js
Database	MongoDB
ODM	Mongoose
API Testing	Postman / Thunder Client
Code Editor	Visual Studio Code
AI Service	Google Gemini AI

The tools used during development are Visual Studio Code, Node.js, npm, MongoDB, Postman and a web browser.

3.2 Hardware Requirements

The FITTRACK backend can be developed and tested on a standard modern computer.

Component	Minimum Requirement
Processor	Intel Core i5 8th Gen or equivalent
RAM	8 GB (16 GB recommended)
Storage	Minimum 1 GB available
Internet	Required for relevant AI/cloud services

3.3 Environment Variables

The following environment variables are used for server configuration, database connection, authentication and Gemini AI integration.

Variable	Purpose
PORT	Server configuration
MONGO_URI	Database connection
JWT_SECRET	Authentication
GEMINI_API_KEY	Gemini AI integration

4. System Architecture

FITTRACK follows a modular MVC-based backend architecture.

4.1 Overall Architecture

Client → Express Server → Routes → Authentication Middleware → Controllers → Services / Mongoose → Gemini AI / MongoDB

4.2 Architecture Components

Client Layer

The client sends requests to the FITTRACK backend and receives JSON responses. Possible clients include a web application, a mobile application, Postman and Thunder Client.

Express Server

The Express server receives API requests and manages communication between the routes, controllers, services and the database.

Route Layer

Routes define the API endpoints available to the client. The major route categories are authentication routes, workout routes and AI routes.

Authentication Middleware

The authentication middleware verifies the JWT token before allowing access to protected APIs.

Controller Layer

Controllers process incoming requests and generate responses.

Service Layer

The service layer contains reusable application services.

Service	Responsibilities
Gemini AI Service	Sends fitness information to Gemini AI, receives AI-generated recommendations and generates fitness insights.
JWT Service	Creates and verifies JWT tokens and manages authenticated user information.
Password Service	Performs password hashing and password verification.

Database Layer

FITTRACK uses MongoDB for storing application information. Mongoose is used for database connection, schema definition, data validation and database operations.

4.3 MVC Architecture

Layer	Description
Model	Manages database information using MongoDB, Mongoose, the User model and the Workout model.
View	Represents the client consuming the API, such as a web application, mobile application, Postman or Thunder Client. The backend returns JSON responses.
Controller	Receives requests, validates input, executes application logic and communicates with the models and services.

5. Database Design

FITTRACK contains two major entities: User and Workout.

5.1 User Entity

Field	Description
_id	Unique user ID
name	User name
email	User email
password	Encrypted password
createdAt	Creation time
updatedAt	Update time

5.2 Workout Entity

Field	Description
_id	Unique workout ID
user	User reference
workoutName	Workout name
category	Workout category
duration	Workout duration
caloriesBurned	Calories burned
workoutDate	Workout date
createdAt	Creation time
updatedAt	Update time

6. Key Features

Feature	Description
Secure Authentication	User registration and login; JWT authentication protects restricted API routes; passwords are handled using bcrypt.js.
Workout Management	Authenticated users can add, view, update and delete workout records.
Workout Search	Workout records can be searched by workout name, category and date.
Google Gemini Integration	Provides personalized workout recommendations based on user information.
AI Fitness Insights	Generates insights using workout statistics such as total workouts, average duration and calories burned.
Centralized Error Handling	Handles errors and validation centrally across the backend.
RESTful API	Follows REST principles and provides structured JSON responses.
Modular Architecture	Separates routes, controllers, services, models, middleware and configuration.

7. User Roles and Authentication

7.1 Roles and Permissions

The primary role in FITTRACK is the authenticated user, who is permitted to perform the following functions.

Function	Permission
Register	Allowed
Login	Allowed
View Profile	Allowed
Add Workout	Allowed
View Workout	Allowed
Search Workout	Allowed
Update Workout	Allowed
Delete Workout	Allowed
AI Recommendation	Allowed
Fitness Insights	Allowed

7.2 Authentication Process

Register → Login → JWT Generated → JWT Sent With Request → Middleware Verification → Protected API Access

If the JWT is missing or invalid, the protected API request is rejected.

7.3 User Flow

Start → Register → Login → Receive JWT → Access Dashboard / APIs → Manage Workout → Search Workout → Request AI Recommendation → View Fitness Insights → End

8. System Usage

After registering, the user can log in using an email and password. A successful login generates a JWT token, which is used to access protected APIs. The authenticated user can:

1. Add a workout.
2. View all workouts.
3. Search workouts.

4. View a particular workout.
5. Update workout details.
6. Delete workouts.
7. Request AI recommendations.
8. Generate fitness insights.

9. AI Integration

FITTRACK uses Google Gemini AI to provide personalized fitness guidance.

9.1 User Inputs

The user can provide age, fitness goal and experience level.

9.2 AI Output

Gemini AI generates the following:

- Personalized workout recommendations.
- Weekly workout suggestions.
- Recommendations according to experience level.
- Motivational guidance.
- Safety-oriented suggestions.

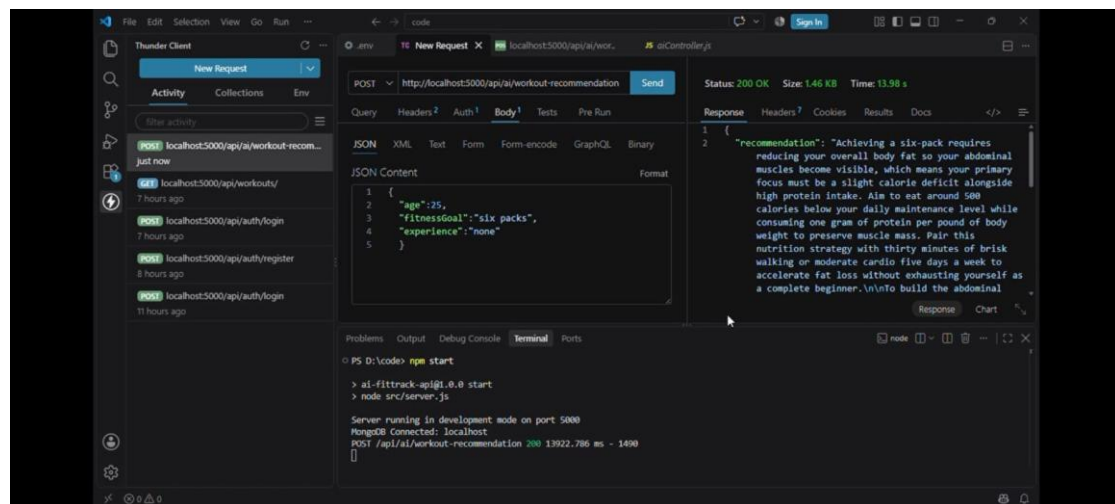


Figure 1: AI-generated Workout Recommendation

9.3 Fitness Insights

The system can also use total workouts, average workout duration and calories burned to generate personalized fitness insights.

9.4 AI Flow

User Information → Age + Fitness Goal + Experience → Gemini AI → Personalized Recommendation

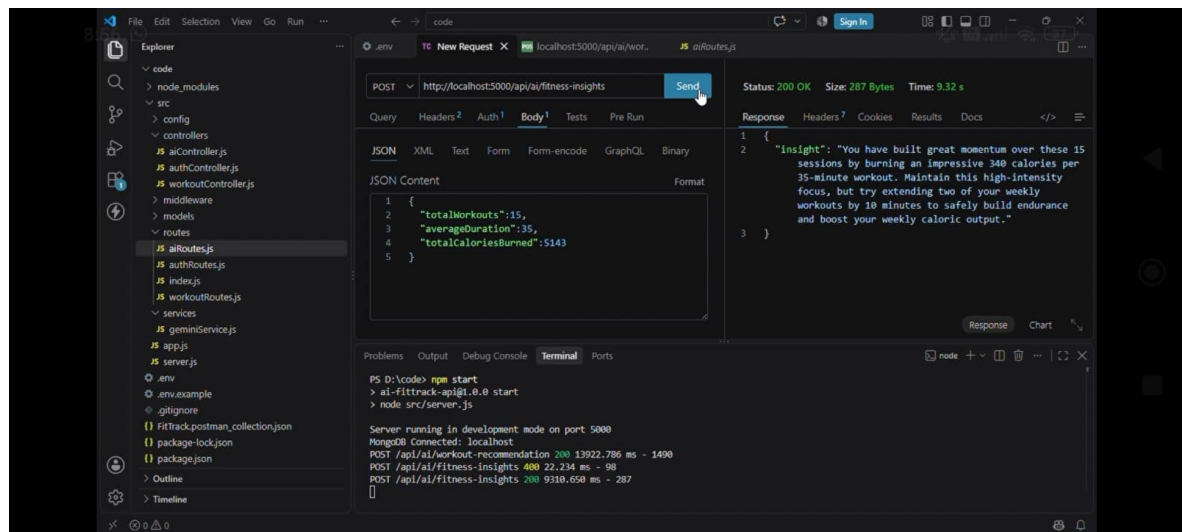


Figure 2 : AI-generated fitness recommendation

10. Project Implementation

10.1 Project Setup and Configuration

The project folder is named AI-FitTrack-API. The project is initialized using npm with the following command:

```
npm init -y
```

The required packages are installed using:

```
npm install express mongoose dotenv bcryptjs jsonwebtoken cors
```

The development dependency is installed using:

```
npm install --save-dev nodemon
```

The environment variables are configured as follows:

```
PORT=YOUR_PORT  
MONGO_URI=YOUR_MONGODB_CONNECTION  
JWT_SECRET=YOUR_SECRET  
GEMINI_API_KEY=YOUR_API_KEY
```

10.2 Backend Development

The backend is divided into the following modules.

Module	Responsibility
Server Configuration	Starts the Express application.
Database Configuration	Connects the application to MongoDB.
Authentication	Handles registration, login, JWT and password protection.
Workout Management	Handles create, read, update, delete and search operations.
AI Recommendation	Communicates with Google Gemini AI.
Middleware	Protects authenticated routes.
Error Handling	Provides centralized API error handling.

10.3 Database Configuration

MongoDB is used as the primary database, with two collections: Users and Workouts. Mongoose provides schemas and validation for these collections.

User Validation

- Name is required.
- Email is unique.
- Password is protected.

Workout Validation

Each workout record contains the workout name, category, duration, calories burned, workout date and a user reference.

10.4 Project Execution

The backend server is started using the configured npm command. Once the server is running, the APIs can be tested using Postman or Thunder Client.

10.5 API Testing

Category	APIs
Authentication APIs	Register User, Login User, Get Profile
Workout APIs	Add Workout, Get All Workouts, Search Workout, Get Workout By ID, Update Workout, Delete Workout

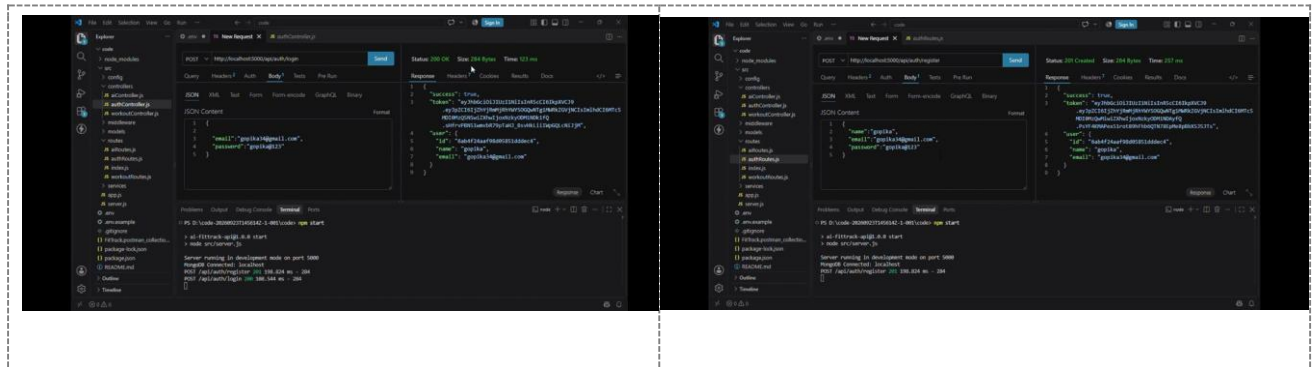


Figure 3: Authentication API testing (Register and Login)

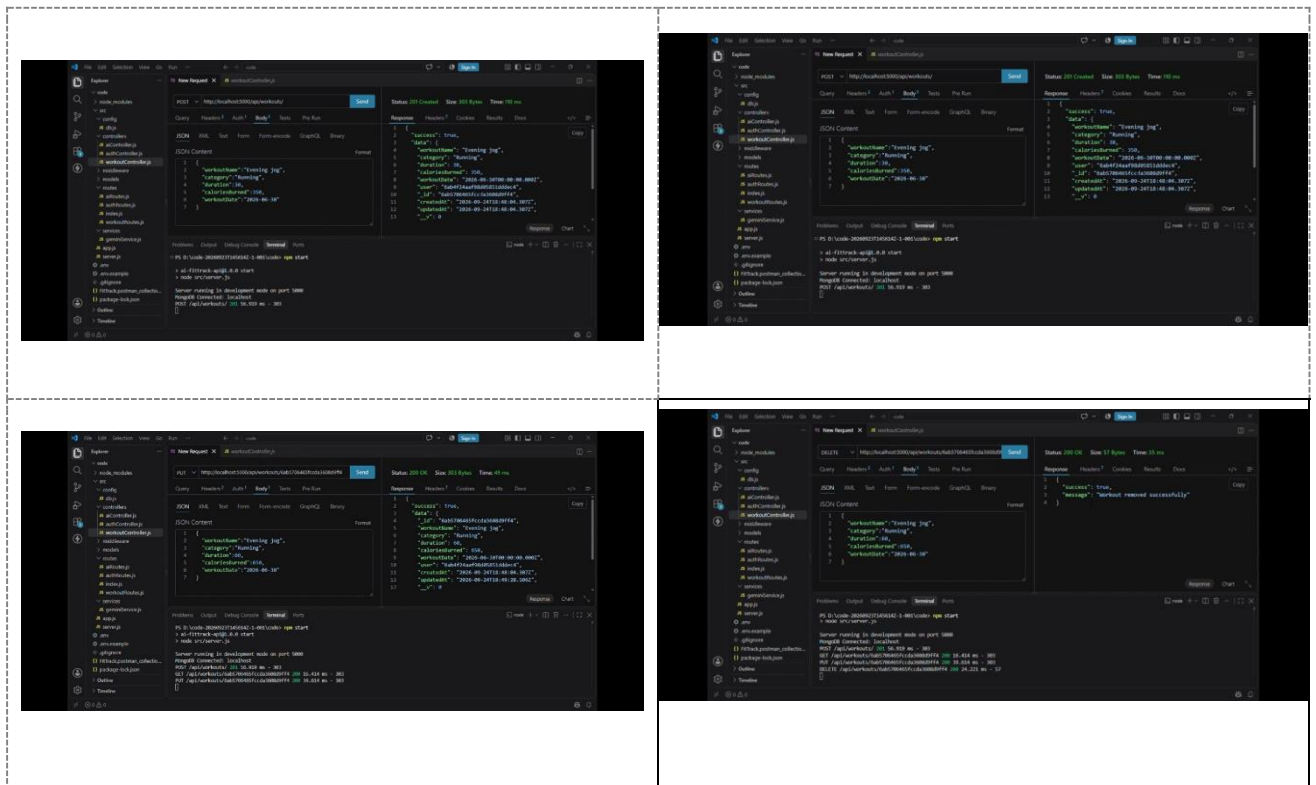


Figure 4: Workout API testing

10.6 Demo Video

Demo video: [https://drive.google.com/file/d/1zfuad8u5R8TkgZ2-LdP5FNvbvUY_zKAG/view?usp=drivesdk]

11. Outcome

11.1 Outcome

FITTRACK provides a centralized solution for managing fitness and workout information. The system reduces the need for manual workout record management and provides an organized method for storing and retrieving fitness data.

11.2 Major Outcomes

- Secure user authentication.
- Centralized workout management.
- Easy workout searching.
- Personalized AI recommendations.
- Fitness performance insights.
- Structured REST APIs.
- Modular backend architecture.
- Secure database access.

12. Conclusion

The project demonstrates how a secure RESTful backend can combine structured fitness tracking with AI-powered assistance. The modular design provides a foundation for future interface development and additional fitness-related capabilities.